

Austin Luk

778-680-2954 | Vancouver, BC | lukaustin0@gmail.com | [linkedin.com/in/austinluk](https://www.linkedin.com/in/austinluk) | github.com/austinluk

EDUCATION

University Of British Columbia

Graduating May 2027

Bachelor of Science in Computer Science

Relevant Coursework: Machine Learning (CPSC 330), Introduction to Artificial Intelligence (CPSC 322) Data Structures and Algorithms (CPSC 221), Software Construction (CPSC 210), Database Systems (CPSC 304),

Langara College

Sept 2023 – April 2025

Associate of Science Computer Science

Activities and societies: Vice-President of Langara Volleyball Club, Langara Computer Science Club (LCC)

EXPERIENCE

Code Ninjas

March 2025 - Current

Instructor

Vancouver, BC

- Developed logical skillset and programming knowledge of children using Scratch, JavaScript, Lua, and C#/Unity.
- Taught over 50 - 100 students coding fundamentals, guiding them using Socratic questioning
- Managed high-volume classroom environments, rapidly diagnosing and resolving technical issues under time pressure.

Headstarter AI

July 2024 – Sept 2024

Software Engineer Fellow

Vancouver, BC

- Designed and deployed cloud-based AI applications using Next.js, OpenAI APIs, and managed databases
- Led and deployed full-stack projects using MVC design patterns through mentoring and feedback from industry leaders. (ex. Google, Meta, Facebook)

PROJECTS

[CourtIQ \(Volleyball Tracking System\)](#) | Python, PyTorch, OpenCV

- Developed a real-time computer vision system to track volleyball player movements and ball trajectory from video input
- **Trained and fine-tuned deep learning models** in PyTorch for object detection and motion tracking
- Built a data processing pipeline to extract positional data and generate gameplay analytics (e.g., player movement, ball speed, rally patterns)
- Visualized tracking outputs and metrics to support performance analysis and gameplay insight

[TaPRO](#) | Gemini API, Next.js, Node.js, Supabase

- Built an AI-powered virtual teaching assistant platform to support university office hours for students and professors.
- Designed data pipelines to store assignments, course materials, and interaction summaries in Supabase for fast retrieval and reuse.
- Implemented optimized retrieval pipelines to reduce API payload size and improve response latency by 40%.

[ThenStep](#) | Next.js, Bun, OpenStreetMap, Google Maps API, Puppeteer

- Developed an advanced map application that allows users to search for both normal driving and walking routes, as well as scenic alternatives.
- Leveraged Puppeteer for web scraping to gather data on safety reports, local meetups, and community events.
- Integrated OpenStreetMap and Google Maps API for dynamic and interactive mapping.

TECHNICAL SKILLS

Machine Learning: PyTorch, NumPy, Pandas, Model Training & Evaluation, Data Preprocessing

Languages: Python, Java, JavaScript, TypeScript, C++, SQL, HTML, CSS,

Frameworks & Libraries: React, NodeJs, Spring, Express, REST APIs, Git, MySQL, MS SQL, MongoDB

DevOps: Amazon Web Services (AWS), Docker, Kubernetes, Azure, CI/CD

Certifications: AWS Certified Cloud Practitioner (In Progress – Expected May 2026)